

important to facilitate encrypted communication and the platform should have authorisation capabilities. It should be secure enough to defend the end user data from all sorts of backdoors, DDoS attacks, website vulnerabilities, malware and even rootkits.

- **Operational expenses:** When evaluating an IoT platform, the end user must consider parameters like pay per node, pay per active device, pay per message, the price of premium features, the price of support, etc.

Let's now look at nine IoT platforms that fulfil all the above requirements and are the best in the business.

Kaa

Kaa is a production-ready, flexible, multi-purpose middleware platform for developing end-to-end IoT solutions, connected applications and smart products. It provides a holistic approach for implementing rich communication, control and interoperation capabilities in connected products and smart devices. It is primarily based on flexible microservices and easily adapts to almost any need and application. It scales from tiny startups to a massive enterprise, and supports advanced deployment models for multi-cloud IoT solutions. Kaa enables effective data management for connected objects and strong back-end infrastructure by providing server and endpoint SDK components. The Kaa platform reduces the associated costs and risks of developing IoT products, while speeding up the time-to-market.

The Kaa IoT platform consists of three main components.

Kaa Server: This is the back-end of the platform and is used to manage tenants, applications, users and devices. It provides all sorts of integration interfaces and administration support.

Kaa extensions: This comprises software modules to improve the overall functionality of the platform.

Endpoint SDK: This is regarded as the library that provides client-side APIs for various Kaa platform features

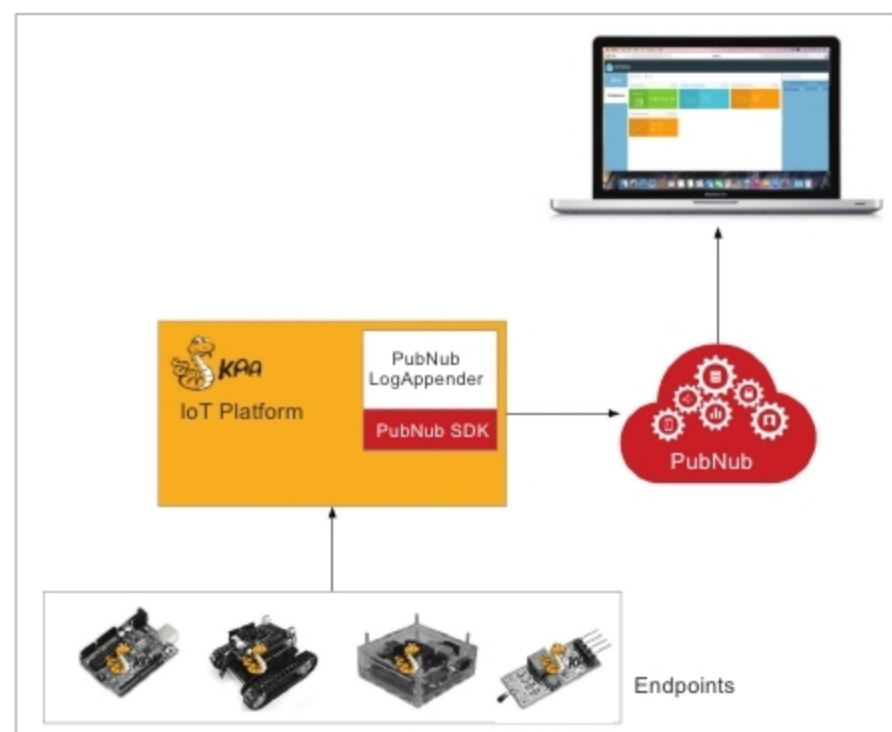


Figure 1: The Kaa IoT platform

and handles all sorts of communication, data marshalling, persistence, etc. It is designed for all sorts of customised client applications to run a wide range of connected devices.

Features

- It is a sophisticated mechanism for the delivery of configurable event messages across connected devices. Events can be unicast or multi-cast.
- Facilitates cross-device interoperability.
- Performs real-time device monitoring, remote device provisioning and configuration.
- Consists of topic-based notification systems to allow end users to deliver messages of any predefined structure to subscribed endpoints.
- Provides updates of operational data like configuration data from server to endpoints.
- Offers a wide range of options when it comes to selecting the network stack for establishing communication between the server and endpoints — Wi-Fi, Ethernet, ZigBee, MQTT, CoAP, XMPP, TCP, HTTP, etc.

Official website: <https://www.kaaproject.org/>

SiteWhere

SiteWhere is an open source IoT application enablement platform that facilitates ingestion, storage, processing and integration of device data at a massive scale. It provides a complete management system to enable end users to cut all per-device charges imposed by SaaS operators. It facilitates connectivity of devices using different protocols like MQTT, SMQP, STOMP, etc, and devices can even be added through self-registration, REST services or in batches. The platform is based on the modern microservices architecture and is designed for reliability, high throughput, low latency and stability.

The platform is licensed under CPAL 1.0 and can be implemented as an on-premise solution with complete freedom and scalability. The platform syncs well with third party integration frameworks like Mule AnyPoint to allow even data to trigger complex interactions, like adding data to Salesforce. It protects information by allowing only limited access to data, using the best user management systems.

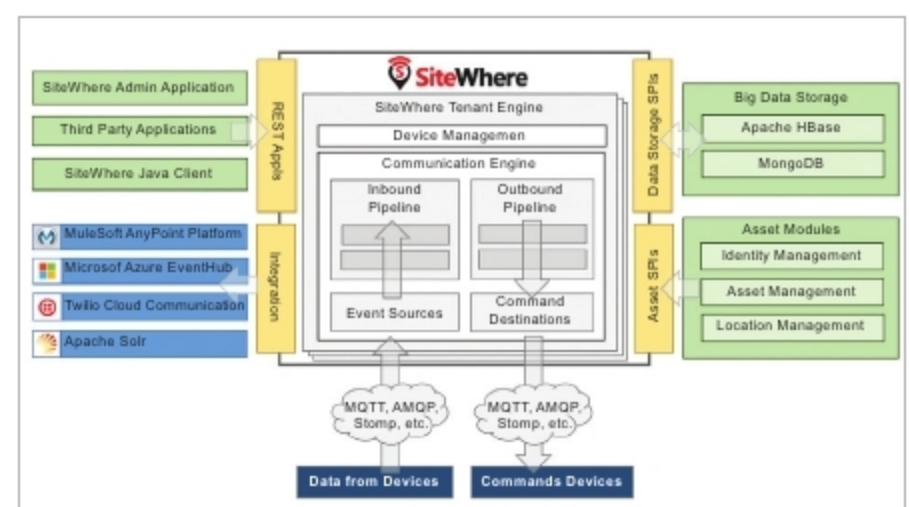


Figure 2: SiteWhere components